

Solve the following exercises

Exercise 1. Identify the language generated by each of the following context-free grammars.

Notes:

Upper case symbols are non-terminals

Lower case symbols are terminals

ε represents an empty character (nothing)

a. $S \rightarrow aaSb$

$S \rightarrow \varepsilon$

b. $S \rightarrow TS$

$S \rightarrow \varepsilon$

$T \rightarrow aTb$

$T \rightarrow \varepsilon$

Exercises 2-5 based on: Sebesta, Robert, Concepts of Programming Languages

Exercise 2. Using the grammar below

$\langle \text{assign} \rangle \rightarrow \langle \text{id} \rangle = \langle \text{expr} \rangle$

$\langle \text{id} \rangle \rightarrow A | B | C$

$\langle \text{expr} \rangle \rightarrow$
 $\langle \text{id} \rangle + \langle \text{expr} \rangle$
 | $\langle \text{id} \rangle * \langle \text{expr} \rangle$
 | $(\langle \text{expr} \rangle)$
 | $\langle \text{id} \rangle$

show a parse tree and a leftmost derivation for each of the following statements:

a. $A = A * (A + (B + A))$

b. $C = B * (A + C * B)$

c. $C = ((A + B) * C)$

Exercise 3. Show that the following grammar is ambiguous:

$\langle S \rangle \rightarrow \langle A \rangle$

$\langle A \rangle \rightarrow \langle A \rangle + \langle A \rangle$

 | $\langle \text{id} \rangle$

$\langle \text{id} \rangle \rightarrow a | b | c$

Exercise 4. Convert the following BNF grammar to EBNF

$\langle \text{assign} \rangle \rightarrow \langle \text{id} \rangle = \langle \text{expr} \rangle$

$\langle \text{id} \rangle \rightarrow A | B | C$

$\langle \text{expr} \rangle \rightarrow$
 $\langle \text{expr} \rangle + \langle \text{expr} \rangle$
 | $\langle \text{expr} \rangle * \langle \text{expr} \rangle$
 | $(\langle \text{expr} \rangle)$
 | $\langle \text{id} \rangle$

Exercise 5. Convert the following EBNF to BNF:

```
<sign> --> + | -  
<comma_digit> --> 0 | 1 | ,  
<comma_integer> --> [<sign>] <comma_digit> {<comma_digit>}
```

Exercise 6. What would be the output of this program if we consider dynamic scoping

```
def function_a ():  
    x = 0  
    y = 4  
    function_b()  
    print (x+y+z)  
  
def function_b ():  
    z = 10  
    y = 9  
    function_c()  
    print (x+y+z)  
  
def function_c ():  
    z = 5  
    x = 6  
    print (x+y+z)  
  
x = 1  
y = 2  
z = 3  
function_a()  
print (x+y+z)
```

Exercise 7. Given the Clite grammar in our slides derive the following program, highlighting if it is or is not part of the language (no syntax error).

```
int main()  
{  
    int i=5;  
    int n=200;  
    while ((i < n/2))  
    {  
        if ((n % 2) == 0)  
            n = n/2;  
        else  
            n = n-11;  
    }  
}
```